

Multiple Choice (10 marks)

1. Higher dietary sodium (salt) intake is generally associated with:
 - (a) Decreased calcium excretion in the urine
 - (b) Increased risk of fracture
 - (c) Decreased dietary calcium absorption
 - (d) Negative calcium balance and bone mineral loss
2. The measurement of a specific biochemical status index for a micronutrient such as vitamin C is most likely to be useful in practice to estimate:
 - (a) The presence or risk of a clinical deficiency, such as clinical scurvy.
 - (b) The overall quality of a person's diet
 - (c) The tissue levels, or body status, of that nutrient.
 - (d) Overall health and freedom from disease.
3. Which of two hormones act on the hypothalamus to increase satiety and hunger, respectively?
 - (a) Leptin and insulin
 - (b) Leptin and ghrelin
 - (c) Ghrelin and leptin
 - (d) Ghrelin and insulin
4. Which of the following statements about glucose metabolism is correct?
 - (a) Glucagon increases the rate of glycolysis
 - (b) Glycolysis requires NADP⁺
 - (c) In glycolysis, glucose is cleaved into two three-carbon products
 - (d) The end-product of glycolysis in red blood cells is pyruvate
5. Which vitamin is required for synthesis of the blood clotting proteins?
 - (a) Vitamin A
 - (b) Vitamin D
 - (c) Vitamin E
 - (d) Vitamin K

True / False (10 marks)

Answer True or False

6. True or False: Acute binge drinking is associated with a condition known as Holiday heart syndrome.
7. True or False: The Western diet is known to support brain health and prevent cognitive decline as we age.
8. True or False: Food allergies usually occur only after an individual has been sensitized by prior exposure to the allergen.
9. True or False: Additional dietary protein may reduce the risk of hip fractures by decreasing bone loss.
10. True or False: Macrobiotic and vegan diets are typically low in sugar, focusing on whole, natural foods.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of acetaldehyde in cancer development and analyze how alcohol consumption can influence the formation of this metabolite in the body.
12. Discuss how acrodermatitis enteropathica, an inborn error of metabolism, leads to zinc deficiency and analyze its effects on health and development.

End of Paper